Tanmay Agarwal

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Summary

Software Engineer with ~2 years experience across industry and research roles specializing in LLM systems (MS CS, Columbia, Dec 2026). Ships full-stack AI products end to end: an LLM document-automation pipeline at Regeneron (750+ docs, 98% accuracy), a 23-tool agentic networking platform (React, TypeScript, Supabase), and a codebase-visualization tool spanning 40+ languages. Co-author of 2 arXiv preprints on QLoRA fine-tuning and LLM agent evaluation. Available full-time January 2027.

Technical Skills

- **Languages**: C, C++, Python, JavaScript, SQL, R, Java, HTML/CSS
- **AI / ML**: Machine Learning, LLMs, Agentic Systems, Transformer Fine-Tuning, RAG, Prompt Engineering, PyTorch, Scikit-learn, XGBoost, CUDA, AWS Bedrock, MCP Protocol, Deep Learning
- **Frameworks & Tools**: React, FastAPI, Node.js, Express, MongoDB, Three.js, WebSockets, REST APIs, Power BI, APScheduler, Deepgram, ElevenLabs, Git, Vite, Jupyter Notebook, Weights & Biases, AWS
- **Systems & Infra**: Distributed Systems, Server-Sent Events (SSE), OAuth 2.0, CI/CD, Linux, Windows
- **Soft Skills**: Problem Solving, Leadership, Critical Thinking, Communication, Adaptability, Collaboration

Education

Columbia University **Aug 2025 - Dec 2026**
M.S., Computer Science *New York, NY*
- **GPA:** 3.68
- **Achievements:** Swami Vivekanand Scholarship for Academic Excellence - Covering 50% of tuition, PDL Fellowship

Vellore Institute of Technology **Aug 2021 - Jun 2025**
B.Tech, Computer Science & Engineering - Data Science Specialization *Vellore, India*
- **GPA:** 8.7

Experience

Stealth Startup under Columbia Build Labs **Sep 2026 - Present**
Foundation Engineer *New York*
- Own the end-to-end AI stack by building a real-time voice agent with ElevenLabs, a persistent user-memory layer with SQL and Appwrite, and an LLM chat assistant with GPT and Claude APIs, providing voice interaction, persistent context, and AI-powered chat in one product
- Shipped 8 merged PRs to production in week one with 73 automated tests, including mutation checks, and closed a CI evidence-gate bypass

Regeneron Pharmaceuticals **May 2026 - Aug 2026**
AI Intern *Tarrytown, NY*
- Implemented an LLM-powered document automation pipeline (Python, AWS, Bitbucket CI/CD, WebSockets) projected to save Regeneron **$95k** annually by eliminating **6+** hours of manual processing per document change control
- Engineered a document extraction and validation system using Claude, RAG retrieval, structured output parsing, and human-in-the-loop review, processing **750+** documents at **98%** accuracy

Columbia University (CRIS LABS) **Sep 2025 - May 2026**
LLM Research Assistant *New York, NY*
- Developed production AI pipelines and dynamically loadable MCP skill modules for industrial fault-detection systems, using FastAPI and CI/CD to deploy LLM-powered components across three active engineering workflows
- Designed and delivered LLM components for adaptive fault detection with **FastAPI, MongoDB, and OAuth 2.0**, expanding the codebase to support over 8K citations in leading engineering journals
- Collaborated within an **NSF-funded lab** led by a National Academy of Engineering member, shipping clean, tested, production-ready Python code on a weekly cadence

Suryaa Chamball Power Limited **Jan 2025 - Aug 2025**
AI / Data Analyst *India*
- Developed end-to-end predictive ML models (**Python, scikit-learn, XGBoost**) for real-time power loss estimation across **13+** sensor stations, lifting forecast accuracy by **18%** over baseline
- Engineered an automated ETL pipeline eliminating **15+** hours of weekly manual work, cutting processing time by **45%** and raising data accuracy by **22%** for downstream model training
- Deployed real-time Power BI dashboards integrating **5 live sensor** streams, enabling proactive anomaly detection and slashing estimated stakeholder response time by **57%**

Projects

Tam - Personal AI Life OS https://mytam.netlify.app **Aug 2026 - Oct 2026**
Columbia University *New York City*

- Architected Tam+, a full-stack AI networking platform (React, TypeScript, Supabase Postgres, Claude) unifying 4 prior apps into 1 codebase with 3 build targets (local-first, multi-tenant SaaS, public demo), spanning 15+ screens, 6 edge functions, and 7 row-level-secured tables
- Engineered a 23-tool agentic assistant (8-step loop cap) and 7 LLM workflows over 37,000+ LinkedIn connections and 24,000+ messages across 4674 conversations, auto-scoring contacts 0-100 and scheduling outreach on 7/30-day cadences into a paced daily queue
- Built the SaaS layer end to end: 5-step self-serve onboarding, 14-day trials, Stripe subscriptions via webhooks, admin-issued access codes, and server-enforced AI limits (150 requests/user/day plus a scope guard), batch-ranking 240 contacts in 6 LLM requests, backed by 16 automated tests

CodeScope AI https://github.com/TanmayAgarwal123/CodeScope-AI-Williamsburg **Jan 2026 - Feb 2026**

- Shipped a full-stack AI platform auto-interpreting and visualizing multi-language codebases at >90% structural accuracy, slashing code-comprehension time by 60% and reducing average dev onboarding from 5 days to under 2
- Leveraged AWS Bedrock LLMs for cross-language logic translation across 40+ programming languages, powering a real-time Q&A layer boosting onboarding efficiency by 40%

Snapgram - Full-Stack Social Media App https://snapgram101.netlify.app **Aug 2025 - Sep 2025**

- Built a full-stack social platform with **8+ core features** (posts, image uploads, likes, saves, profiles, explore, search, infinite scroll) using React, TypeScript, Appwrite, and Tailwind CSS, deployed on Netlify. Combining plus points of both snapchat and instagram
- Cut redundant API calls by **23%** with TanStack Query caching and optimistic updates, bringing like/save response time under **100 ms**
- Reached a **90+ Lighthouse performance score** through lazy-loaded images, paginated infinite scroll, and **300 ms** debounced search, keeping the feed smooth at **1k+ posts**

Publications

- Beyond Static Leaderboards: Predictive Validity for the Evaluation of LLM Agents. https://arxiv.org/abs/2606.19704 **Jun 2026**
- Internalizing Tool Knowledge in Small Language Models via QLoRA Fine-Tuning. https://arxiv.org/abs/2605.17774 **May 2026**
- Designing Energy-Aware Scheduling and Task Allocation Algorithms for Online Reinforcement Learning Applications in Cloud Environments. https://ieeexplore.ieee.org/document/10803910 **Dec 2024**

Leadership & Achievements

- **Technical Project Head, Soft Computing Research Society, VIT Vellore:**Mentored 50+ students and drove completion of 30+ end-to-end projects in ML, Web, and App development
- **Google Devfest Organizer:**Organized Google DevFest 2026 in collaboration with NVIDIA, attracting 700+ registrations and welcoming 200+ attendees
- **Core Member, Google Developer Student Clubs (GDSC), VIT:**Organized 3 technical workshops averaging 60+ attendees each
- **Smart India Hackathon (SIH):**Advanced to National Level out of 1M+ participants nationwide
- **Competitive Coding:**802+ problems on CodeChef · 307+ problems on LeetCode
- **Freelance Developer:**Delivered 4 end-to-end client projects spanning web and AI development
- **Music Artist:**5 Self-released originals on Spotify, Apple Music, Amazon Music, YouTube Music
- **International Science Olympiads:**20+ awards across mathematics, science, and computing